

ERIC (CHEN-CHIEN) CHANG

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SUMMARY

- Backend / Data Infrastructure Engineer specializing in **performance tuning, distributed systems, and open-source data platforms**.
- Core stack: **Go, Java, Rust, TypeScript, PostgreSQL, Kafka, Docker, Kubernetes**. Delivered measurable wins including first-byte latency **100s to 5s**, WebUI load time **60s to 1s**, and simulation jobs **2hr to 1min**.

OPEN SOURCE CONTRIBUTIONS

Apache Kafka

2024 - Present

*Apache Kafka Contributor. Tech Stack: **Java, JVM***

[GitHub](#)

- Collaborating on Kafka Streams **KIP-770** input-buffer work ([PR #22458](#)) to add `input.buffer.max.bytes`, bounding buffering by bytes instead of record count for safer memory control under variable record sizes and partition assignments.
- Fixed **JVM container awareness** in producer performance tests and offset commit-before-partition-switch correctness ([PR #21584](#)).

Apache Gravitino

2024 - Present

*Apache Gravitino Committer. Tech Stack: **Java, Python, Docker, Kubernetes***

[GitHub](#)

- Added PostgreSQL/MySQL integration-test container management and reproducible Docker image workflows; reduced Hive image size by **420MB** and ported Python client partition APIs.

Apache DataFusion and DataFusion Comet

2025 - Present

*Contributor. Tech Stack: **Rust, Scala, Apache Spark***

[GitHub](#)

- Implemented Spark-compatible `str_to_map` in Rust ([PR #20120](#)) and wired Comet's native SparkStrToMap UDF with SQL file coverage ([PR #3654](#)).

redis/rueidis

Oct 2023 - Present

*Contributor. Tech Stack: **Go, Redis***

[PR #395](#)

- Implemented the go-redis API Adapter for teams migrating from go-redis.

SELECTED PROJECTS

TWFoundry

2026

*Hybrid cloud / homelab Taiwan operations data platform. Tech Stack: **Airflow, Kafka, ClickHouse, Cloudflare R2, Workers/Pages, Docker, TypeScript***

[GitHub](#)

- Designed a hybrid cloud / homelab data platform for Taiwan operations, using homelab Airflow and Kafka for scheduled ingestion and backfills, Cloudflare R2 for raw and projection artifact storage, ClickHouse for OLAP analytics, and Cloudflare Workers / Pages for public edge APIs and UI delivery.
- Built a materialized edge serving model where Workers / Pages Functions read partitioned R2 artifacts and expose replayable map/timeline projection APIs without falling back to homelab on cache misses.

- Modeled TDX transit observations in ClickHouse across service date, time slot, route, direction, vehicle, freshness, route progress, and stop context to support route density, data-quality, and headway / bunching analysis.

EXPERIENCE

Lawsnote

January 2024 - December 2024

Backend Software Engineer. Tech Stack: **Typescript, NodeJS, Docker, MongoDB, Prometheus, Grafana** Taipei, Taiwan

- Implemented **HTTP streaming** for CPU-intensive computation results, reducing first-byte latency from **100s** to **5s**.
- Built a **MinIO events** and **Elasticsearch** monitoring path to detect missing data proactively and improve pipeline reliability.
- Profiled and fixed a **NAPI-rs** performance issue with **Pyroscope**, improving Node.js native package execution.
- Built an **LLM service** for extracting information from ambiguous regulatory text and automated government act crawling, re-crawl, and re-parse workflows.

Mediatek (Contract)

February 2022 - March 2023

Backend Software Engineer. Tech Stack: **Go, Linux, Docker, PostgreSQL** Hsinchu, Taiwan

- Maintained a low-power simulation and benchmarking data warehouse, improving WebUI page load time from **60s** to **1s** with **PostgreSQL index-only scans**: [Blog Post](#).
- Used **Pyroscope eBPF** to identify import bottlenecks, then applied **PostgreSQL bulk import** to reduce simulation job duration from **2hr** to **1min**.
- Designed a message-queue-based microservice architecture and operated a **5-node** service cluster for horizontally scalable long-running jobs.
- Reduced Go service memory consumption by **50%** using **sync.Pool**: [Blog Post](#).
- Built integration-test fixture factories and maintained observability with **Prometheus, Grafana, OpenTelemetry**, and **Pyroscope**.

National Chen-Kung University

September 2017 - July 2019

Research Assistant. Tech Stack: **Python, C**

Tainan, Taiwan

- Conducted FDM 3D printing research across firmware, filament experiments, and hardware/software integration.
- Built experimental tooling with **Python, C, MicroPython**, and **Arduino**.

EDUCATION

National Taiwan Ocean University, Electrical Engineering coursework (Drop Out)

2013 - 2017

SKILLS

Languages: Go, Java, Rust, TypeScript, Python, C

Data Warehouse / Lakehouse: PostgreSQL, ClickHouse, Hive, StarRocks

Streaming / Processing: Kafka, Kafka Streams, Flink, Spark, SQL, dbt

Runtime / Orchestration: Airflow, Docker, Kubernetes, Linux

Observability / Data Quality: Prometheus, Grafana, OpenTelemetry, Pyroscope, Great Expectations

DevOps / Applied AI: Git, GitHub Actions, LLM service integration